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Deep learning identification of coronary artery disease from bilateral finger photoplethysmography sensing: A proof-of-concept study

Sadaf Iqbal, Sharad Agarwal, Ian Purcell, Alan Murray, Jaume Bacardit, [John Allen](#)

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Abstract

Background: A proof-of-concept study assessing a novel approach to identify patients with coronary artery disease (CAD) using deep learning analysis of bilateral-site photoplethysmography (PPG) waveforms ("DL-PPG").

Methodology: DL-PPG was studied in 37 participants (with 21 having CAD). Scalogram 'spectral' images were derived from right and left index finger PPG measurements collected using a 3-phase protocol (baseline, unilateral arm pressure cuff occlusion, reactive hyperaemia flush). Artificial Intelligence (AI) analysis, namely deep learning, was employed for scalogram image classification using a Convolutional Neural Network (CNN, "GoogLeNet"), with classification performance obtained using 10-fold stratified cross validation (CV). A conventional machine learning (ML) classifier (K-nearest neighbour, K-NN, K = 9) was also evaluated for comparison with the CNN deep learning methodology. Blood samples were also collected giving 2 biochemical biomarkers of endothelial function. Test sensitivities, specificities, accuracies, and Kappa statistics were determined.

Results: DL-PPG sensitivity was 80.9 % (95% CI, 78.6–83.0), specificity 87.7% (85.5–89.7), accuracy 83.8 % (82.2–85.3), and Kappa 0.68 (0.65–0.71). Comparative K-NN ML performance was 69.4% (95% CI, 68.7–70.1), 37.5% (36.7–38.2), 53.9% (53.3–54.4), and 0.069 (0.058–0.079), respectively. No differences between patients and controls were found for the biochemical biomarkers of endothelial function.

Conclusion: Substantial overall agreement was found between DL-PPG classification and CAD angiography, with DL-PPG performance clearly better than for a conventional ML technique. Our deep learning classification approach, using only basic pre-processing of the PPG pulse waveforms before classification, could offer significant benefits for the diagnosis of CAD in a variety of clinical settings needing low-cost portable and easy-to-use diagnostics.

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Keywords

- Artificial intelligence
- Coronary artery disease
- Deep learning
- Heart
- Machine learning
- Photoplethysmography
- Pulse
- Endothelial function

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KW - Deep learning

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KW - Machine learning

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